

Perceiver IO Model for Efficient Vibration Signal Compression and Anomaly Detection

Guillaume Lebonvallet¹, Luis A. Salazar-Zendeja², Faicel Hnaïen³, Hichem Snoussi⁴, and Brice Nélain⁵

Abstract—This research presents a deep learning model for vibration signal compression and anomaly detection, using Perceiver IO—an attention-based architecture—within a railway industry case study. The model is lightweight and optimized for deployment on energy- and memory-constrained sensors, making it ideal for embedded applications. Unlike traditional methods, it performs signal compression and anomaly detection simultaneously, offering an efficient and unified solution applicable across industries. To avoid the need for separate classification models or heavy architectures, contrastive learning is used to improve feature extraction. This allows the learned representations to be easily separable via a simple linear function, enabling effective anomaly detection while preserving signal reconstruction quality. The model is trained on a large vibration signal dataset using a weighted loss function combining mean squared error (mse), cosine similarity, and contrastive loss. Contrastive learning plays a key role in clustering normal and abnormal signals, achieving near-perfect classification accuracy. Compression performance is evaluated with metrics such as peak signal-to-noise ratio (PSNR), cosine similarity, and compression ratio. The model achieves a PSNR of 21.87 while maintaining a much smaller architecture. The approach shows strong potential for real-time industrial deployment, ensuring high accuracy and efficiency in resource-constrained environments.

Index Terms—Anomaly detection, attention mechanisms, contrastive learning, embedded computing, neural network compression, signal processing.

I. INTRODUCTION

VIBRATION analysis is a critical field in engineering, widely used to monitor the health of mechanical systems and detect anomalies that could indicate potential failures. By identifying early signs of defects, predictive maintenance strategies can be implemented to enhance reliability, minimize downtime, and prevent costly repairs. The effectiveness of monitoring vibration patterns for predictive maintenance has been well-documented [1]. The nonintrusive nature of vibration analysis makes it an attractive solution across various

industries, including marine engineering [2], petroleum [3], railways [4], and aerospace [5].

With the increasing complexity of machinery and the surge in sensor-generated data, efficient processing and representation of vibrational signals have become key challenges. Traditional vibration analysis methods struggle with noisy high-dimensional data, requiring models that can effectively capture essential features while reducing redundancy [6]. Compression techniques not only mitigate storage and transmission constraints, but also enhance downstream tasks, such as fault classification and anomaly detection [7]. These advances are particularly valuable for real-time industrial applications, where data efficiency is crucial.

To address these challenges, advanced machine learning techniques have become powerful tools for vibration analysis, extracting meaningful patterns from large sensor data for efficient compression and fault detection. Compact yet informative signal representation reduces data bottlenecks while preserving diagnostic information. In addition, adaptive learning helps distinguish normal variations from true anomalies, enhancing fault detection in complex industrial settings.

Machine learning and deep learning techniques provide several advantages in vibration analysis: the ability to process large volumes of data efficiently, identify complex patterns, and adapt to evolving conditions [8], [9]. Over the years, numerous models have been proposed for fault detection and diagnosis [10], [11]. Approaches such as K -means clustering [12], convolutional neural networks (CNNs) [13], and long short-term memory networks [14] have demonstrated high accuracy in detecting faults. More recently, deep autoencoders combined with ensemble learning [15] have shown robustness under noisy conditions.

Vibration signal compression plays a vital role in condition monitoring; however, it presents significant challenges due to the requirement to efficiently store and transmit data while preserving critical diagnostic information. Key challenges include balancing compression efficiency with signal integrity and managing computational complexity in real-time applications. Addressing these challenges, various advanced methods have been proposed. For instance, Oltean et al. [16] explored tailored compression techniques suited for aircraft engine monitoring, ensuring high fidelity in signal reconstruction. Meanwhile, Guo and Tse [17] introduced an approach leveraging optimal ensemble empirical mode decomposition to enhance compression performance while preserving key signal characteristics. Furthermore, Yin et al. [18] integrated

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Guillaume Lebonvallet, Luis A. Salazar-Zendeja, Faicel Hnaïen, and Hichem Snoussi are with the LIST3N (Computer Science and Digital Society), Université de Technologie de Troyes, Troyes 10004, France, and also with Eut+ Data Science Institute, European Union, Troyes 10004, France (e-mail: guillaume.lebonvallet@utt.fr; luis.salazar_zendeja@utt.fr; faicel.hnaïen@utt.fr; hichem.snoussi@utt.fr).

Brice Nélain is with the Innovation Team, VOSSLOH, 69100 Villeurbanne, France (e-mail: brice.nelain@vossloh.com).

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encryption into the compression process, providing a dual advantage of data security and lightweight storage, which is critical for remote monitoring systems. Collectively, these studies highlight the ongoing advancements in vibration signal compression, addressing challenges related to efficiency, accuracy, and security in industrial applications.

Many real-world applications require these models to be embedded on sensors, which impose strict memory and energy constraints. Traditional deep learning models can be computationally expensive, making them impractical for edge-based condition monitoring. To ensure feasible deployment in industrial settings, lightweight models that balance accuracy with efficiency are essential. Embedded fault detection systems, such as those presented in [19], demonstrate the effectiveness of compact machine learning architectures tailored for edge devices. In addition, compressive sensing techniques designed for wireless sensor networks [20] offer promising solutions for reducing data transmission overhead while maintaining signal integrity. A broader overview of AI models in the Industrial IoT applications [21] further emphasizes the necessity of optimized algorithms that can operate within the constraints of embedded environments.

To the best of our knowledge, existing literature addresses compression and anomaly detection as distinct tasks, which can result in redundant computations and increased resource consumption, particularly problematic in real-time industrial applications. Moreover, the constraints of embedded environments, including limited memory and power availability, make it challenging to deploy separate models efficiently. To address these challenges, we propose a unified Perceiver IO-based architecture that integrates vibration signal compression and anomaly detection within a single framework. Our key contributions are threefold.

- 1) We develop a latent space embedding that extracts compact yet informative representations of vibration data, achieving high compression efficiency without compromising critical diagnostic features.
- 2) We design an adaptive anomaly detection mechanism that selectively focuses on relevant patterns, filtering out noise and enhancing fault diagnosis reliability in industrial environments.
- 3) We propose a unified, lightweight framework that efficiently combines compression and anomaly detection, making it ideal for resource-constrained embedded systems. This integration ensures our approach is well-suited for large-scale industrial applications where real-time processing and accuracy are paramount.

This article is organized as follows. Section II reviews related work on vibration analysis, fault detection, and signal compression; Section III presents the architecture of the Perceiver IO model, the data preprocessing steps, the attention mechanisms for compression and anomaly detection, and the implementation of contrastive learning to improve feature separability; Section IV describes the research context, the dataset, and the specific training of our model; the results are discussed in Section V; and the article concludes with Section VI, which covers the conclusions and future prospects.

II. RELATED WORK

Deep learning techniques have demonstrated considerable promise in vibration signal analysis, particularly for fault detection in mechanical systems. In recent years, several studies have explored the use of deep learning models to predict system health from vibration signals. For example, Lin et al. [22] evaluated various deep learning neural networks for surface roughness prediction, using vibration signal analysis. Their findings showed that deep learning models, particularly CNNs, can successfully extract features from vibration signals and provide accurate predictions. Similarly, Chen and Lee [23] investigated how CNNs can enhance vibration signal analysis for applications like fault diagnosis and surface roughness estimation. They utilize both 1-D and 2-D CNN models to process raw vibration data and time–frequency representations, improving predictive accuracy. Their approach integrates optimization techniques to refine CNN performance, demonstrating its effectiveness in detecting faults and assessing machining conditions.

For bearing fault diagnosis, Wang et al. [24] fused vibroacoustic data with a 1-D CNN network, showing significant improvements in diagnostic accuracy compared to traditional methods. Another study [25] proposed a distributed approach for bearing fault diagnosis based on vibration signals. This method leveraged multiple sensors to collect vibration data, improving the fault diagnosis process by reducing the impact of sensor placement and environmental noise.

More recently, the use of explainable artificial intelligence (XAI) has gained traction in vibration signal analysis for fault diagnosis. Chen and Lee [26] explored the use of XAI for bearing fault diagnosis, where they employed machine learning models with interpretability features. This approach allows for a deeper understanding of the decision-making process, which is particularly beneficial in critical applications, like industrial maintenance.

Signal compression has been a significant focus in vibration analysis, as it helps reduce storage and transmission requirements while preserving diagnostic information. Biagetti et al. [27] analyzed lossless compression algorithms for wireless sensor networks, such as EMG sensors, highlighting trade-offs between compression ratio and signal fidelity—crucial for real-time vibration monitoring. Xin and Fan [28] introduced a lossless method for medical image compression using big data mining to achieve high efficiency while preserving data integrity. Though focused on medical data, their approach is also applicable to vibration signals due to similar requirements for data preservation.

The review of existing methods demonstrates that, while significant advancements have been made in vibration signal compression and fault diagnosis, many challenges remain. Traditional approaches often address compression and anomaly detection separately, leading to inefficiencies and increased computational demands, particularly in real-time industrial applications. Furthermore, many existing techniques struggle to balance compression efficiency with the preservation of critical fault detection features, especially when dealing with noisy or high-dimensional data. These challenges are further exacerbated in embedded environments, where strict memory

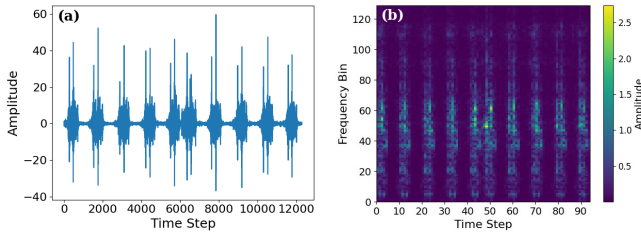


Fig. 1. (a) Time domain signal. (b) STFT representation.

and energy constraints limit the feasibility of complex models. To address these limitations, this work proposes a unified Perceiver IO-based model that integrates both efficient compression and anomaly detection while remaining lightweight and computationally efficient for resource-constrained edge devices.

III. PROPOSED METHODOLOGY

Monitoring vibration signals is essential for identifying faults in mechanical systems, yet the high dimensionality and inherent noise in raw data often present significant challenges. Traditional models frequently struggle to process such signals efficiently without compromising critical fault-related information. To address these limitations, we introduce a model based on Perceiver IO [29], known for its capability to manage diverse, high-dimensional data with remarkable efficiency.

Building on the strengths of Perceiver IO, our proposed model leverages a latent-space representation to significantly reduce computational requirements and memory footprint, while preserving essential signal structures. The domain-agnostic attention mechanism enables flexible handling of arbitrary input and output formats, ensuring adaptability to various vibration-based fault detection scenarios. This design is particularly effective in isolating meaningful features from noise, enhancing the clarity and reliability of fault detection.

A key enhancement of this model is its ability to seamlessly integrate compression and anomaly detection using a single unified framework. By employing attention-driven mechanisms, the model dynamically identifies fault-indicative patterns and suppresses background interference. Moreover, we incorporate contrastive learning to refine the anomaly detection process, eliminating the need for an additional decision-making model. These innovations streamline the architecture, improving both efficiency and sensitivity to faults.

Sections III-A–III-D detail the application of this enhanced model to vibration signal compression and contrastive learning-driven anomaly detection, following initial preprocessing of time-domain signals. This unified approach achieves acceptable performance in terms of both computational efficiency and fault detection accuracy, positioning it as an advanced solution for industrial condition monitoring.

A. Preprocessing: Short-Time Fourier Transform

In this study, we apply the short-time Fourier transform (STFT) to our vibration signal analysis. STFT is advantageous for examining nonstationary signals, as it provides a

time–frequency representation that captures transient events and frequency variations, which are crucial for accurate fault detection in machinery. This method enhances the detection of localized anomalies in vibration data, thereby improving diagnostic precision [30].

The STFT process involves segmenting the vibration signal into overlapping windows and performing a Fourier Transform on each segment. This approach maintains both time and frequency information, allowing a detailed analysis of how signal components evolve over time. By selecting appropriate window sizes, a balance between time and frequency resolution can be achieved, making STFT a versatile tool for analyzing complex vibrational signals in various engineering applications. Fig. 1(a) shows an example of a time-domain signal and Fig. 1(b) shows a time–frequency signal.

B. Architecture of the Proposed Perceiver IO

The overall framework, as shown in Fig. 2, consists of three main components: vibration signal compression, attention-driven anomaly detection, and vibration signal reconstruction. Initially, the input vibration signals undergo preprocessing using STFT, as described in Section III-A. The STFT representations are initially transformed into a latent space through two fundamental steps: 1) a cross-attention block that interacts between the input signals and the latent space, and 2) a series of self-attention processes within the latent space itself. This interaction not only captures the complex relationships within the data but also significantly reduces dimensionality while preserving the essential features needed for accurate representation. This module functions as the encoder in the Perceiver IO model.

Once the compressed representation is obtained, it is directed toward two distinct modules for further processing. The first module is dedicated to anomaly detection, employing a straightforward linear function based on principal component analysis (PCA). This approach assists in identifying deviations from normal operational patterns by analyzing the principal components of the latent representation. The reconstruction module (decoder) takes center stage, leveraging a positional encoding mechanism to reconstruct the STFT representation of the signal from the compressed representation with a cross-attention block.

For each attention module, there is a multilayer perceptron (MLP) and normalization layers. The MLP consists of two fully connected layers with a GELU activation in between, expanding the output to a higher dimension H and then returning it back to its original shape. The normalization layers center and reduce the data with a learned mean and standard deviation for each input value. The MLP introduces nonlinearity and enhances the expressiveness of the model, whereas the normalization layers stabilize and accelerate the training process.

The final output includes both the reconstructed signal and an anomaly score, enabling real-time fault detection. This architecture ensures an optimal balance between efficient data compression and accurate fault identification.

The modular design (encoder, decoder, and anomaly detector) is particularly well-suited for embedded environments. In

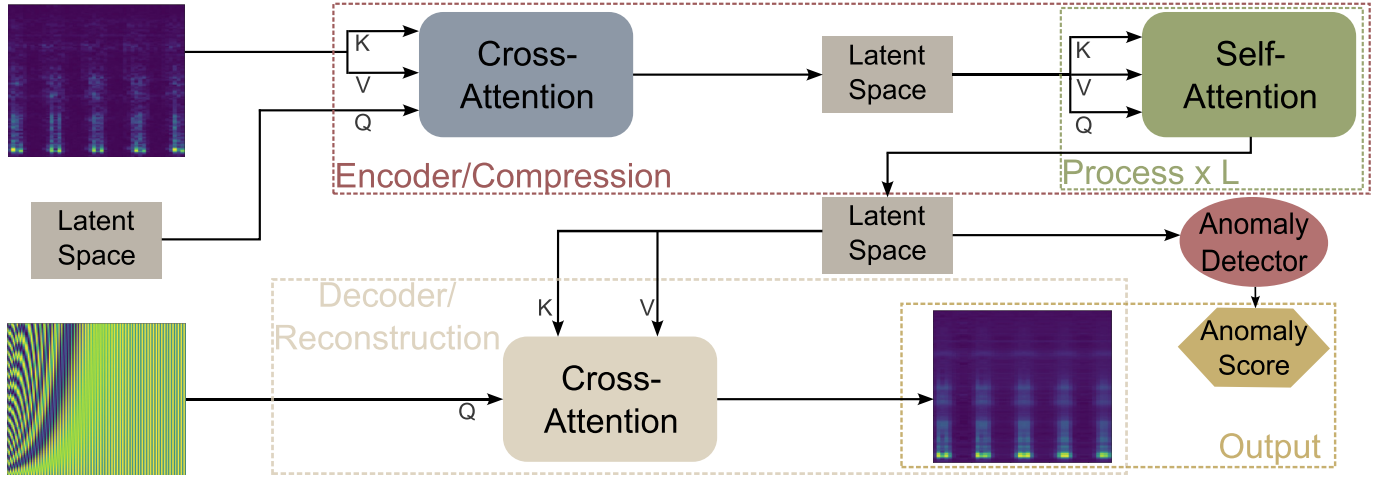


Fig. 2. Perceiver IO architecture for compression and anomaly detection.

practice, only the encoder and anomaly detector are deployed on the edge device, allowing for efficient, on-device processing with minimal computational overhead. The decoder, which is more resource-intensive, is applied downstream after transmission of the compressed representation, enabling optional signal reconstruction for verification, analysis, or archival purposes. This separation of responsibilities ensures low-latency anomaly detection while maintaining the ability to reconstruct signals when needed.

C. Attention Internal: Variable-Sized Inputs

The attention module in Perceiver IO operates in a series of steps, with a design that ensures the length of the key-value input does not affect the fixed size of the output.

The attention mechanism of the encoder begins by projecting the latent representation and the signal data into three matrices: queries (Q), keys (K), and values (V). These projections are linear transformations to a common feature dimension F using learned weight matrices

$$Z \in \mathbb{R}^{M \times C}, X \in \mathbb{R}^{N \times D} \quad (1)$$

$$Q = ZW_Q, W_Q \in \mathbb{R}^{C \times F} \text{ and } Q \in \mathbb{R}^{M \times F} \quad (2)$$

$$K = XW_K, W_K \in \mathbb{R}^{D \times F} \text{ and } K \in \mathbb{R}^{N \times F} \quad (3)$$

$$V = XW_V, W_V \in \mathbb{R}^{D \times F} \text{ and } V \in \mathbb{R}^{N \times F}. \quad (4)$$

Z is the latent representation and X is the input STFT of arbitrary length N . Notice that the learned weight matrices (W_Q , W_K , and W_V) do not depend on N , the length of the input (and neither M , the length of the latent representation). This ensures the mechanism is agnostic to the input size.

The attention scores are calculated as the dot product between the query (Q , projection of the latent) and the transposed key (K^T , projection of the input STFT), scaled by $(F)^{1/2}$ to stabilize gradients. Mathematically

$$A = \frac{QK^T}{\sqrt{F}}, A \in \mathbb{R}^{M \times N}. \quad (5)$$

The resulting attention score matrix A has a shape of (M, N) , indicating that each of the M latent queries attends to all

N input STFT time elements. Since A depends on M (size of the latent representation) and N (input length), it adapts dynamically to variable-sized inputs.

The attention scores are normalized using the Softmax function along the input length dimension (N). This step ensures that the attention weights across all N input elements for each query sum to 1, regardless of the input size. The normalized attention weights are used to compute a weighted sum of the value matrix (V , other projection of the input STFT) and linearly projected on the latent space. This step produces the output of the attention module

$$O = \text{Softmax}(A)VW_O, O \in \mathbb{R}^{M \times C} \text{ and } W_O \in \mathbb{R}^{F \times C}. \quad (6)$$

The key insight is that the shape of O is the same as the given latent representation. The length L of the input affects the intermediate attention computation but does not propagate to the output. This ensures that the final output size remains consistent, irrespective of the input length.

At no point in this pipeline do the learned weight matrices (W_Q , W_K , W_V , and W_O) depend on the input length N (and the latent representation length M). This independence allows the Perceiver IO model to handle inputs of arbitrary size while maintaining a consistent, scalable output. This design is ideal for tasks involving variable-length data, such as the case study presented in Section IV.

In the decoder section of the model, the roles of the inputs are reversed: the latent representations are used as the key and value, whereas the variable-sized output encoding serves as the query. Since the attention mechanism in the decoder mirrors that of the encoder, the resulting output is a matrix of variable length, aligning with the dimensions of the original input STFT.

D. Contrastive Learning

Contrastive learning is a machine learning technique that emphasizes learning representations by comparing data samples to differentiate between similar and dissimilar pairs. The core principle involves pulling together the representations of similar or related data points, known as positive pairs, while

pushing apart the representations of dissimilar or unrelated data points, referred to as negative pairs. Triplet loss, also known as large margin nearest neighbor (LMNN) [31], is a crucial loss function in this context. It aims to organize data points within an embedding space by ensuring that similar data points are closer together and dissimilar ones are farther apart, maintaining a specified margin between different classes or clusters. This method normally operates on triplets of samples: an anchor, a positive example from the same class as the anchor, and a negative example from a different class. The loss function encourages the model to learn embeddings where the distance between the anchor and positive is minimized, and the distance between the anchor and negative is maximized beyond a certain margin. This margin ensures a meaningful separation between clusters.

In our approach, the contrastive loss assesses the similarity between the two latent matrices, utilizing only a selected portion (i.e., a specified number of columns) of each matrix for the computation. This approach is based on the understanding that the characteristics crucial for anomaly detection are often distinct from those essential for reconstructing the STFT. By segregating these characteristics, the model minimizes the adverse impact of the contrastive loss on the reconstruction process. This ensures that the vital features for reconstruction remain unaffected by the contrastive component, thereby enhancing the overall performance and accuracy of the model. When pairs of vectors originate from the same cluster, the contrastive loss aims to minimize the distance between them, promoting similarity within clusters. Conversely, for pairs from different clusters, the contrastive loss enforces a margin constraint, ensuring these vectors remain sufficiently separated in the latent space. This separation is vital to ensure accurate anomaly detection without the dedicated classifier. Our implementation introduces a notable modification to the traditional LMNN or triplet loss framework [31]. Unlike the original formulation, where the margin is dynamically calculated, we fix the margin to a constant value. This adjustment prevents excessive influence on the reconstruction process, maintaining a balance that allows the model to learn meaningful cluster distinctions without compromising its ability to accurately reconstruct the input data. The implementation of contrastive learning within this framework is detailed in Section IV-C. This tailored approach enhances the robustness and overall effectiveness of the model in anomaly detection tasks.

IV. EXPERIMENTAL SETUP

A. Case Study on Railway Monitoring

This research was developed within the context of railway infrastructure monitoring, specifically addressing the challenges of switch and crossing system surveillance [32], [33]. In our industrial context, the railway monitoring system implements a hierarchical data flow architecture designed to optimize railway infrastructure monitoring while addressing the constraints of remote deployment and real-time decision-making. Vibration signals are first captured by autonomous IoT sensors (located at switch and crossing systems) equipped with energy harvesting capabilities and embedded processing

algorithms. Rather than transmitting raw data, these sensors perform local processing to extract metadata and detect anomalies at the source. This strategy significantly reduces bandwidth usage and enables immediate responses to potential issues.

The processed information is then transmitted to modular Virtual Observation Gateway units, which act as data concentrators. These units aggregate input from multiple sensors and conduct intermediate analysis before forwarding critical insights to centralized servers. At the highest level of the architecture, advanced AI algorithms perform global analysis across the network to predict maintenance needs and coordinate proactive interventions.

This distributed processing approach is essential, as it reduces communication overhead in remote railway environments, supports real-time local decision-making for safety-critical events, minimizes data storage requirements, and enhances system resilience by distributing intelligence throughout the network.

While the broader railway monitoring system incorporates a full hierarchical data flow, our work specifically focuses on the development of lightweight AI models for signal compression and anomaly detection capable of operating directly on embedded sensor nodes.

B. Dataset Description

To develop and validate the model, a dataset of vibration signals was collected under various operating conditions in the railway network. Due to data protection policies, some details of the data collection process are omitted. The signals were acquired using accelerometers mounted on multiple railway switches and crossing systems. The dataset consists of 13 907 time-series recordings, with sampling frequencies ranging from 3100 to 3350 Hz. Of these, 5000 are anomalous, and each recording lasts between 3 and 26 s.

The signals were recorded under different conditions, including variations in train types, number of wagons, speeds, and ambient temperature. To preprocess the data, we applied a simple trimming procedure to remove noise from the beginning and end of each signal. Furthermore, STFT, as detailed in Section III-A, was used for further feature extraction.

C. Training of the Model

To ensure the separability of the anomalies over the normal data, a supervised contrastive learning is introduced along the reconstruction training. The goal of the contrastive part is to expand the margin between the two clusters to allow a quick discrimination of samples without needing a full decision model. Algorithm 1 shows the training loop within each epoch. Each training sample is associated with another sample to form pairs for the contrastive loss calculus. Our implementation uses a list of shuffled indices to ensure random and diverse pairings. The signal pairs remain fixed during the whole training process. The code iterates over these shuffled indices, selecting the original sample and its paired counterpart from the dataset. For every pair of inputs (x_1, x_2) , the encoder f maps these inputs into their latent representations z_1 and z_2 (L2-3). The decoder

Algorithm 1 Training Epoch

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1: for each  $(x_1, x_2)$  do
2:    $z_1 \leftarrow f(x_1)$ 
3:    $z_2 \leftarrow f(x_2)$ 
4:    $\hat{x}_1 \leftarrow f^{-1}(z_1)$ 
5:    $L_R \leftarrow 1 - \frac{1}{N} \frac{x_1 \hat{x}_1^\top}{\|x_1\| \cdot \|\hat{x}_1\|} + \frac{1}{ND} \sum (x_1 - \hat{x}_1)^2$ 
6:    $L_C \leftarrow \begin{cases} \frac{1}{ND} \sum (\bar{z}_1 - \bar{z}_2)^2 & \text{if same cluster} \\ \min(\text{MARGIN} - \frac{1}{ND} \sum (\bar{z}_1 - \bar{z}_2)^2, 0) & \text{else} \end{cases}$ 
7:   if batch is finished or no remaining data then
8:      $L_G \leftarrow \lambda_1 L_R + \lambda_2 L_C$ 
9:     Backpropagate  $L_G$  and update model weights
10:  end if
11: end for

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f^{-1} then reconstructs x_1 as $\hat{x}_1$ (L4). The reconstruction loss L_R measures the quality of the reconstruction by combining a cosine similarity term and a mean squared error (mse) term (L5). The cosine similarity assesses the alignment in the feature space, whereas the mse ensures the pixel-wise accuracy. As we want to maximize the cosine similarity between the original signal and the reconstruction, the cosine similarity term is defined as one minus the actual cosine similarity value, allowing us to minimize the reconstruction loss toward zero.

In parallel, the contrastive loss assesses the similarity between selected parts of the two latent matrices. These submatrices, called $\bar{z}_1$ and $\bar{z}_2$, are compared based on their clustering (L6). For pairs from the same cluster, L_C minimizes the distance between the latent vectors. For pairs from different clusters, it enforces the given fixed margin to maintain separation in the latent space.

At the end of each batch, or when no data remains, the global loss L_G —a linear combination of L_R and L_C (L8)—is backpropagated to update the weights of the model (L9). The coefficients λ in this combination control the relative influence of each loss term, allowing for fine-tuning of their impact on training.

After completing the training loop for an epoch, the model is evaluated on the validation data to compute the validation losses. The validation data is processed similar to the training data, but without updating the model parameters. The validation losses are averaged and used to compute the overall validation loss.

V. RESULTS AND DISCUSSION

Our final goal is to achieve the implementation of the model into an embedded target with constrained computation capabilities and memory capacity. Therefore, we have reduced the architecture to accommodate our use case without losing too much performance on the metrics. The latent matrix has a shape of eight rows (M) and 32 columns (C). The cross-attention module of the encoder has a feature dimension (F) of 128 channels, and its MLP has a hidden layer (H) of 256 neurons. We choose to include one process block with a feature size of 64 channels and a hidden size of 64 neurons. The decoder has a feature dimension of 128 channels and a hidden layer of 512 neurons. All attention modules are single-headed

to reduce the memory footprint of the model. In the following experiments, models were trained using the Adam optimizer with a learning rate of 0.001 over 50 epochs and a batch size of 256 signals. The fixed margin used inside the contrastive loss is set to 1. To ensure robust evaluation, we employed a threefold cross-validation strategy for training all models. In each fold, 9271 signals were used for training and 4636 for testing. Both training and testing sets were balanced to contain an equal number of normal and anomalous signals.

A. Performance of CS and Deep Learning Models

Compressed sensing (CS) is a signal acquisition paradigm that allows the reconstruction of sparse or compressible signals from significantly fewer measurements than required by the Nyquist theorem [34]. In vibration signal processing, CS has been used to reduce data transmission and storage demands in wireless sensor networks by leveraging the inherent sparsity of mechanical vibration signals in suitable transform domains [18], [20].

To establish a baseline for comparison with our proposed Perceiver IO model, we implemented the CS method by [20], which combines an improved K -SVD dictionary learning algorithm with sparse adaptive matching pursuit (SAMP) for automatic sparsity estimation. We selected this method because it is specifically designed for signal compression in wireless sensor networks, which closely aligns with our use case. Our implementation used 128 dictionary atoms, trained over 50 iterations. SAMP was configured to run for 800 reconstruction iterations with a tolerance of $1e^{-4}$.

We selected 727 signals from our dataset corresponding to a single train type operating at a fixed speed, in order to simulate an almost homogeneous dataset where the signals are highly similar to one another. Each signal was trimmed to contain exactly 8000 samples. The method was evaluated at a compression ratio of 0.5, meaning that 4000 measurements were acquired from 8000-sample signals. This setup allowed us to compare its performance against our proposed deep learning-based compression approach.

To ensure both methods operated at the same compression ratio, we increased the latent space dimension of the Perceiver IO model from eight rows and 32 columns to 32 rows and 125 columns.

This comparison focuses on signal compression. In the case of the Perceiver IO, Algorithm 1 computes only the reconstruction loss L_R (and $\lambda_1 = 1$), whereas the contrastive loss L_C , which is part of the full training objective, is omitted in this context.

In the case of the K -SVD algorithm, the output is a reconstructed signal in the time domain. In contrast, our model produces the STFT representation of the signal. To enable a comparison between the two approaches, we applied the real-time phase gradient heap integration (RTPGHI) phase reconstruction algorithm [35] and a classic inverse STFT algorithm to convert the output of our model back into the time domain. Table I reports the mse and cosine similarity between the original and reconstructed signals in both the time domain and the time–frequency domain for the K -SVD algorithm and our proposed approach. Fig. 3 shows examples

TABLE I
SUMMARY OF TRAINING METRICS

	Time Domain Metrics		Time-Frequency Domain Metrics	
	K-SVD	Proposed Approach	K-SVD	Proposed Approach
MSE	4.960 ± 0.113	7.984 ± 0.183	0.019 ± 0.001	0.011 ± 0.000
Cosine Similarity	0.186 ± 0.000	0.000 ± 0.003	0.585 ± 0.026	0.803 ± 0.003

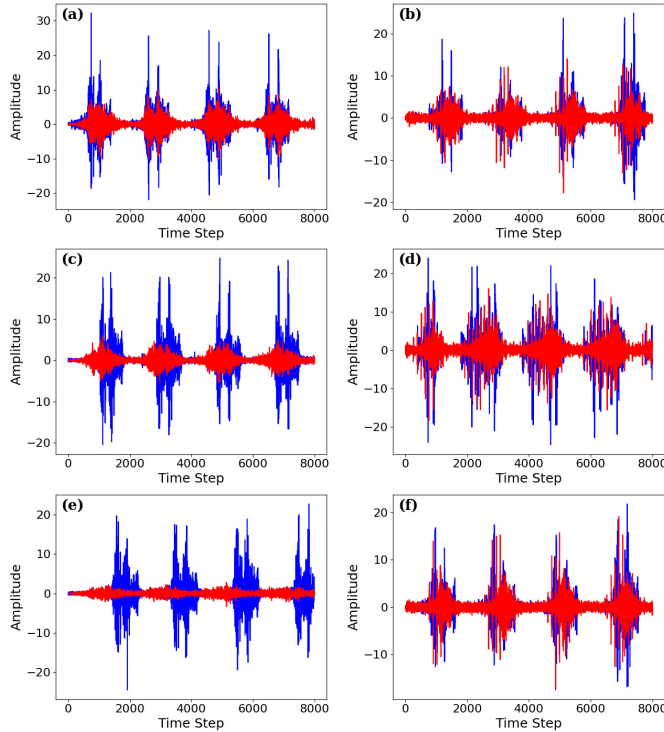


Fig. 3. (a), (c), and (e) Signal reconstructions using the K -SVD algorithm. (b), (d), and (f) Signal reconstructions using the Perceiver IO algorithm.

of the reconstructed time-domain signals. The plots on the left correspond to the K -SVD reconstructions, while those on the right show the outputs from the Perceiver IO model. For each example, the original and reconstructed signals are plotted together to visually assess reconstruction quality.

The time-domain metrics presented in Table I suggest that the K -SVD reconstructions are significantly better than those produced by our model. Notably, the cosine similarity is shown to be near zero. However, the issue lies in the fact that these metrics do not accurately represent the similarity between the reconstruction and the original signal. The use of the RTPGHI algorithm for temporal reconstruction introduces phase shifts that degrade both metrics. For example, a 90° phase shift reduces cosine similarity to zero, while a 180° phase shift maximizes mse. These distortions undermine the validity of time-domain comparisons.

To provide a more accurate assessment of the reconstruction quality of both models, time-frequency metrics are also calculated. In this context, the mse represents the accurate reconstruction of the intensity of each frequency component

at a given time, thus relating to the amplitude of the temporal signal and the fidelity of the spectral content. Conversely, the cosine similarity evaluates the preservation of the overall shape of the temporal signal. Both metrics indicate better performance for our model, which aligns with the visual comparison of the reconstruction samples shown in Fig. 3. K -SVD can reconstruct with a good level of fidelity [see Fig. 3(a)], but it notably struggles to reconstruct signals involving time shifts [see Fig. 3(c) and (e)]. This limitation is common among CS algorithms, which typically rely on fixed dictionaries and assume sparsity in a certain basis. Such assumptions make it difficult for these methods to handle heterogeneous datasets with diverse signal variations, including temporal shifts. Furthermore, as signal length increases, the complexity of accurate reconstruction grows, often leading to degraded performance. Due to these challenges, we restricted our dataset to a pseudohomogeneous set of signals to ensure fair and meaningful comparisons between models. For the following trainings, we use the full dataset available, which includes signals from seven different train types traveling at speeds between 90 and 110 km/h. We observed that while the method performs well, handling a significantly greater diversity of signal patterns may require increasing the size of the latent space and, correspondingly, the overall model. This limitation reflects a common trade-off in AI models between complexity and generalization.

In both models, the peaks corresponding to axle crossings are not well reconstructed. However, our model exhibits artifacts induced by the phase reconstruction algorithm [see Fig. 3(b), (d), and (f)]. These artifacts could potentially be mitigated by fine-tuning the parameters of the STFT, notably reducing the step length. This can lead to improved reconstruction fidelity and reduced phase-related distortions, but will raise the memory footprint.

B. Results on Contrastive Learning

The global loss function, which is a linear combination of normalized metrics, was used to control the influence of each metric during backpropagation. The values of λ_1 and λ_2 are initially chosen arbitrarily. These hyperparameters control the trade-off between optimizing two competing objectives: signal reconstruction and contrastive learning. Increasing one relative to the other prioritizes the corresponding objective during training. In our experiments, we fix $\lambda_1 = 1$, that is, a 100% influence on both the mse and cosine similarity losses, and vary λ_2 to identify a value that effectively separates the classes (normal versus anomaly) while preserving the quality of the STFT reconstruction. For simplicity, we refer to λ_2 as the contrastive weight.

Table II presents a summary of evaluation metrics computed for a series of our proposed models trained with varying contrastive weights. Each row corresponds to a distinct model configuration, where the contrastive weight is incremented from 0.00 to 0.45. The columns report the mean and standard deviation of the performance metrics across the folds, including the mse loss, cosine similarity loss, and contrastive loss. In addition, classification performance is represented by

TABLE II
SUMMARY OF LOSSES AND F1 SCORES ACROSS CONTRASTIVE WEIGHTS

Contrastive Weight	MSE Loss	Cosine Similarity Loss	Contrastive Loss	Accuracy	Normal F1-Score	Anomaly F1-Score
0.00	0.0178 ± 0.0006	0.8140 ± 0.0042	0.9792 ± 0.1755	0.4926 ± 0.2710	0.5262 ± 0.2680	0.5144 ± 0.2492
0.05	0.0210 ± 0.0016	0.7905 ± 0.0068	0.0373 ± 0.0141	0.9981 ± 0.0008	0.9984 ± 0.0004	0.9953 ± 0.0029
0.10	0.0385 ± 0.0027	0.6267 ± 0.0123	0.0075 ± 0.0046	0.9992 ± 0.0006	0.9994 ± 0.0005	0.9986 ± 0.0016
0.15	0.0400 ± 0.0013	0.6215 ± 0.0073	0.0072 ± 0.0033	0.9994 ± 0.0005	0.9991 ± 0.0007	0.9995 ± 0.0003
0.20	0.0402 ± 0.0007	0.6192 ± 0.0106	0.0040 ± 0.0030	0.9999 ± 0.0001	0.9998 ± 0.0001	0.9998 ± 0.0001
0.25	0.0413 ± 0.0010	0.6028 ± 0.0063	0.0096 ± 0.0078	0.9990 ± 0.0010	0.9987 ± 0.0014	0.9991 ± 0.0008
0.30	0.0386 ± 0.0006	0.6261 ± 0.0018	0.0015 ± 0.0008	0.9997 ± 0.0001	0.9997 ± 0.0000	0.9997 ± 0.0002
0.35	0.0395 ± 0.0017	0.6107 ± 0.0025	0.0021 ± 0.0006	0.9996 ± 0.0002	0.9995 ± 0.0003	0.9996 ± 0.0002
0.40	0.0409 ± 0.0007	0.6129 ± 0.0047	0.0057 ± 0.0035	0.9986 ± 0.0009	0.9984 ± 0.0014	0.9985 ± 0.0006
0.45	0.0367 ± 0.0024	0.6238 ± 0.0290	0.1502 ± 0.2089	0.9822 ± 0.0240	0.9854 ± 0.0191	0.9766 ± 0.0320

TABLE III
CLASSIFICATION REPORT AND TRAINING METRICS

		No Contrastive		Contrastive		Partial Contrastive	
		Normal	Anomaly	Normal	Anomaly	Normal	Anomaly
Classification Report	Precision	0.43 ± 0.24	0.24 ± 0.17	0.998 ± 0.001	0.999 ± 0.000	0.992 ± 0.002	0.997 ± 0.000
	Recall	0.29 ± 0.19	0.38 ± 0.23	0.999 ± 0.000	0.999 ± 0.000	0.995 ± 0.001	0.995 ± 0.001
	F1-Score	0.35 ± 0.22	0.30 ± 0.19	0.998 ± 0.001	0.999 ± 0.000	0.994 ± 0.001	0.996 ± 0.000
	Support	8907	5000	8907	5000	8907	5000
Training Metrics	MSE	0.017 ± 0.000		0.040 ± 0.003		0.036 ± 0.002	
	Cosine Similarity	0.813 ± 0.012		0.621 ± 0.014		0.647 ± 0.025	
	Contrastive	0.840 ± 0.049		0.058 ± 0.076		0.025 ± 0.019	

the accuracy and the $F1$ -score for the normal and anomaly classes.

Metrics presented in this table indicate that all configurations, excluding the first two and the last one, are similar in results. The first configuration with no contrastive learning gives us the baseline reconstruction metrics (0.0178 for mse and 81.40% of cosine similarity). The classification accuracy and $F1$ -scores for both normal and anomaly detection improve significantly with the contrastive weight, reaching near-perfect values around weights of 0.20 and above. Notably, at a weight of 0.20, the model achieves an accuracy of 99.99%, with $F1$ -scores exceeding 0.999 for both classes, while the reconstruction metrics (mse and cosine similarity) stabilize to approximately 0.04 for mse and 61% for cosine similarity, indicating minimal compromise in reconstruction quality. Beyond the 0.20 weight, while some configurations like 0.30 and 0.35 also show excellent performance, they do not surpass the 0.20 configuration in terms of overall balance and consistency across all metrics. Therefore, the 0.20 contrastive weight is chosen for its optimal balance between reconstruction quality and classification accuracy, demonstrating the most robust and reliable performance.

To ensure a lightweight and efficient overall model, we avoid incorporating additional complex models for anomaly detection and instead utilize a simple linear classifier. Each signal is classified based on the first PCA component derived from the latent space as shown in Fig. 2. This component is compared against a decision threshold to determine the classification. This threshold is computed by first calculating the mean of the first PCA component for the normal

signals and the mean of the first PCA component for the anomalous signals, and then taking the average of these two means. Compared to heavier classifiers, such as support vector machines (SVMs) and random forests, this approach offers significant computational advantages. SVMs often need complex kernel functions and can become resource-intensive with high-dimensional latent spaces. Random forests, while powerful, involve multiple decision trees, which increase both memory usage and inference time. In contrast, our method reduces overhead by focusing on a single linear decision boundary, making it highly efficient for our real-time and resource-constrained use case.

To isolate the contributions of contrastive learning and the dual-module architecture from other architectural variations, we compare our proposed method against a basic Perceiver IO model that shares the same underlying structure. Table III presents classification and reconstruction metrics for the same model trained with three different loss functions. The first loss does not incorporate contrastive learning ($\lambda_2 = 0$). The second and third models include contrastive learning with $\lambda_2 = 0.2$, but differ in the input used to compute the contrastive loss L_C . The second applies contrastive learning across the entire latent matrix; that is, the contrastive loss L_C is computed using all 8×32 values of the latent matrix. The third, referred to as partial contrastive learning, applies contrastive learning only on the first eight columns of the latent matrix—specifically, a submatrix of size 8×8 from the full 8×32 latent matrix is used to compute L_C .

The model without contrastive loss demonstrates poor classification performance. The mean precision, recall, and

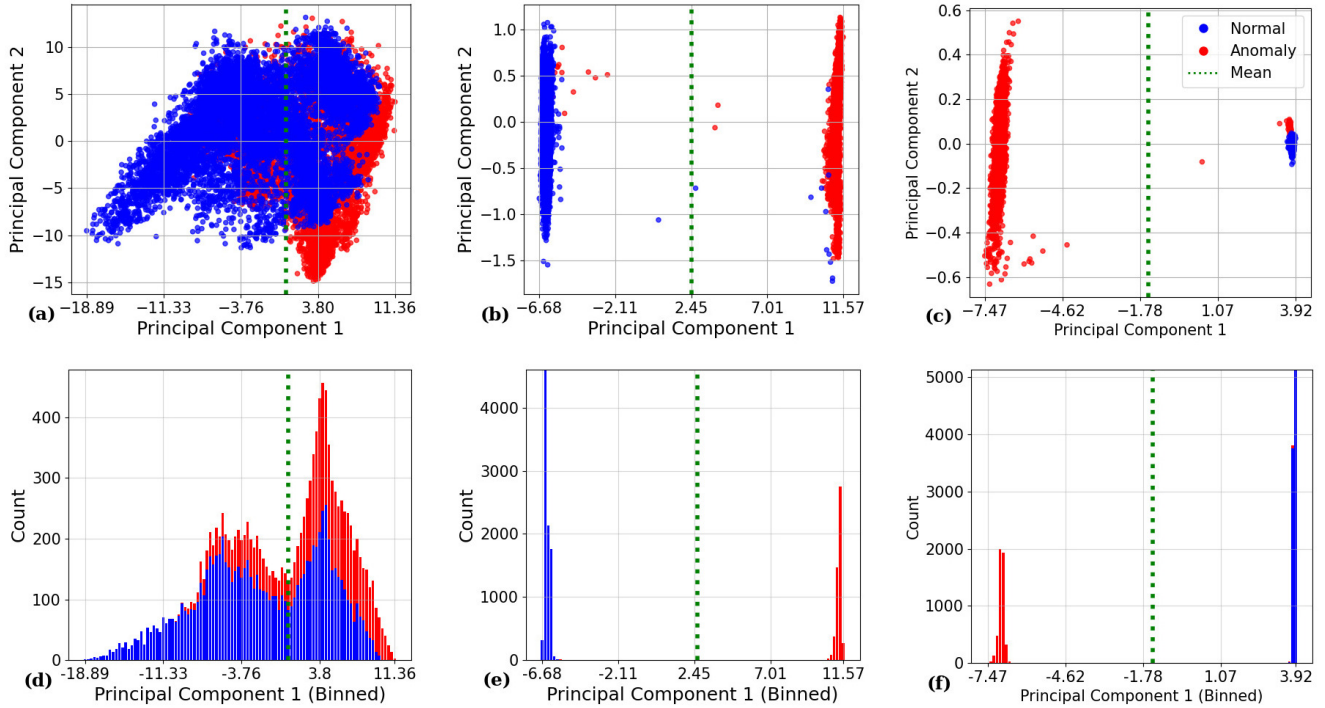


Fig. 4. Projection of the first two PCA components (a) without contrastive learning, (b) with contrastive learning, and (c) with partial contrastive learning. Histogram of the first PCA component (d) without contrastive learning, (e) with contrastive learning, and (f) with partial contrastive learning.

$F1$ -score for the anomalous class are particularly low (0.24, 0.38, and 0.30), indicating that many anomalies are misclassified. These results suggest that the extracted features from the first PCA component are not well-separated, making anomaly detection difficult for the simple detection function. This lack of separation is clearly seen in Fig. 4.

Fig. 4 illustrates the distribution of samples in our dataset projected onto the first two principal components (top row), alongside histograms of the projections onto the first principal component (bottom row). Each visualization corresponds to the model obtained from the average fold of the respective configuration.

As shown in Fig. 4(a), the PCA scatter plot reveals no distinct clustering between normal and anomalous signals. Consistently, the histogram in Fig. 4(d) demonstrates that the threshold (green line) does not effectively separate these two classes when using it as a linear anomaly detection function.

In contrast, the application of contrastive learning markedly enhances classification performance, achieving near-perfect precision and recall values of 0.999 for anomalous signals. This improvement is corroborated visually in Fig. 4(b) and (e), where normal and anomalous signals are clearly segregated along the principal component axis.

Partial contrastive learning, depicted in Fig. 4(c) and (f), yields slightly lower classification metrics compared to full contrastive learning. Nevertheless, it delivers notable improvements in reconstruction quality, with an increase from 62.1% to 64.7% in cosine similarity, as we can see in Table III. Fig. 5 illustrates that reconstructions produced by the partial con-

trastive learning model (bottom row) exhibit superior fidelity compared to those generated by the full contrastive learning model (third row). Moreover, introducing contrastive learning adversely affects reconstruction quality when compared to models trained without it (second row). Given that the classification performance remains largely comparable between the models, with both contrastive learning models misclassifying an average of 50 samples, this trade-off is deemed beneficial for the intended application.

Compression performance was evaluated using key metrics, including the compression ratio, which reflects the size of the compressed data relative to the original. In addition, we considered the peak signal-to-noise ratio (PSNR), an important metric in image, audio, and video compression. Our model achieved a PSNR of 20.46 with a compression ratio of 88. In comparison, the model built in [29], with 20 million parameters, achieved a PSNR of 26.97. Despite our model not achieving the same quality of reconstruction, it is 80 times smaller, demonstrating the efficiency of the Perceiver IO architecture even with smaller models and flexible, nonfixed-size signals.

The use of contrastive learning proves crucial in enhancing the feature separability, which enables the model to cluster the signals effectively, facilitating a simple yet highly accurate anomaly detection process even in the presence of imbalanced classes with a controlled reduction of the compression quality. The strong generalization capabilities of the model demonstrate its effectiveness not only in the railway industry case study but also for potential applications in other domains.

TABLE IV
MODEL SIZE WITH RESPECT TO ITS PARAMETERS

	Baseline	Model A	Model B	Model C	Model D	Model E	Model F	Model G	Model H	Model I	Model J	Model K
Latent (M)	8	4	16									
Latent (C)	32	16	64									
Encoder (F)	128			64	256							
Encoder (H)	256			128	512							
Decoder (F)	128					64	256					
Decoder (H)	512					256	1024					
Heads	1							3				
Processes (L)	1								0	3		
Process (F)	64										32	128
Process (H)	64										32	128
Encoder (KB)	314.89	240.52	463.62	183.30	578.06	314.89	314.89	775.69	264.97	414.73	290.31	364.04
Decoder (KB)	645.13	628.62	678.15	645.13	645.13	323.34	1288.71	875.49	645.13	645.13	645.13	645.13
Total (KB)	960.02	869.14	1141.78	828.43	1223.18	638.22	1603.60	1651.18	910.10	1059.86	935.44	1009.17

*A blank space represent a repeated value from the 'Baseline' column.

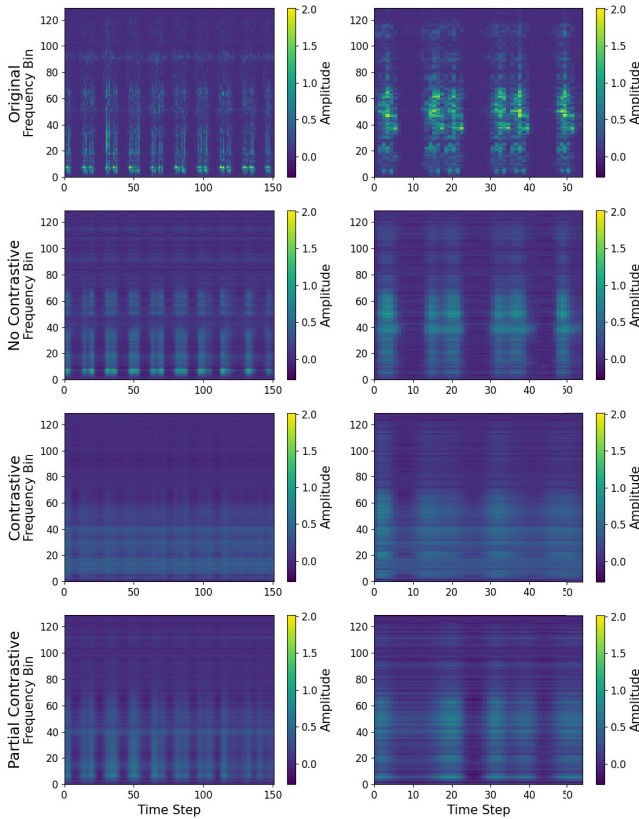


Fig. 5. STFT reconstructions from different models for two sample signals.

C. Embedded Environments

Table IV presents several variants of the model with different configurations compared to the baseline. Although some models (A, C, E, H, and J) reduce the model size and others expand it, finding a balance between these extremes is crucial for embedded environments. In models A and B, the latent matrix size determines the compression ratio, with smaller shapes yielding higher ratios. Models C, D, and H–K adjust the encoder size, whereas E and F modify the decoder. In our use case, only part of the model runs on an embedded target. Given Perceiver IO’s asymmetrical architecture, we

can strategically balance computation and size between the encoder and decoder to minimize the embedded component. Model G demonstrates our choice of single-headed attention, which significantly reduces the model size. Since we prioritize a compact encoder over the decoder, using only one attention head leads to a smaller model, as the MLPs in the encoder are smaller than those in the decoder.

To showcase the embedding capabilities of our model, we have trained a streamlined variant that incorporates parameters from models A, C, E, and H. The new model has a compact architecture, with the encoder measuring 106.01 kB and the decoder at 307.63 kB, resulting in a total model size of 413.64 kB. This model achieved an mse of 0.026 and a cosine similarity of 72.7%. In addition, it produced a PSNR of 21.87. For reference, although not previously reported, the larger model without contrastive learning achieved a higher PSNR of 23.72, indicating improved reconstruction quality due to a larger model capacity. This optimized model is designed to fit within an embedded target while maintaining performance levels that are comparable to larger, less compact alternatives. By demonstrating these capabilities, we affirm the model’s potential for deployment in resource-constrained environments without sacrificing too much accuracy or efficacy.

VI. CONCLUSION AND FUTURE WORK

Our Perceiver IO-based model excels in vibration analysis and fault detection through the attention mechanism, which isolates critical features and filters out noise. The query-based output mechanism enhances precision in identifying anomalies, such as wear or misalignment, in extensive and noisy datasets. The scalability of the model and its contrastive capabilities improve fault detection accuracy and adaptability across various industrial domains. In addition, integrating compression and anomaly detection into a unified framework reduces data dimensionality, accelerates computations, and lowers memory requirements, making it ideal for real-time monitoring and predictive maintenance applications. This combination of features establishes our model as a robust and efficient tool for real-world vibration analysis and fault diagnosis.

In the future, we plan to test our model on an embedded target, specifically utilizing the Arduino Nano 33 BLE Sense Rev2, paired with an ADXL355 3-D axis accelerometer. This development board features 1 MB of flash memory and 256 kB of RAM, which is ample for storing the model weights as well as the code required for signal acquisition, inference, and wireless transmission. We estimate that the model's inference requires approximately 2 million floating-point operations for an input of size (100, 129), corresponding to a temporal signal of roughly 12 000 samples. This suggests that most microcontrollers equipped with a floating-point unit (FPU), such as the nRF52840 integrated within the proposed hardware platform (operating at 64 MHz), could perform inference in under 1 s. This estimation is based on the number of all mathematical operations needed by the different layers of our model. However, the actual computational load may be higher depending on the hardware target and the implementation of the necessary mathematical functions, particularly the exponential and hyperbolic tangent functions. For the same input size, the inference memory footprint, which is the sum of the sizes of all intermediate matrices, is less than 16 kB. We are actively working on further reducing the model's size and computational cost by optimizing its weights. By merging some of the matrices, it is possible to decrease the model's size without affecting its output. These optimizations are inspired by the multihead latent (MLA) attention layer [36]. We will also work on the effect of quantization on the performance of the model, both for the model weights and the STFT intensity values. Reducing the quantification of the STFT amplitudes reduces the size of the input data, which is important in our use case, but may also strongly affect the quality of the temporal reconstructions. In addition, our method may be unsuitable for use cases involving very short signals or requiring extremely fast inference.

Our goal is to expand the applicability of our model by incorporating additional use cases, such as degradation estimation. To achieve this, we aim to further reduce the model's size by more accurately balancing reconstruction and contrastive loss through the implementation of adaptive weights. This optimization will allow us to ensure that the model is compact enough to be fully embedded within the target hardware.

Furthermore, the capability for online training or cooperative learning will enable us to adapt the model dynamically based on incoming data, thereby enhancing its performance and relevance in real-time applications. By refining our approach, we hope to leverage the embedded system's capabilities fully while delivering a robust and versatile solution.

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Guillaume Lebonvallet received the master's degree in optimization and systems safety and the engineering degree in networks and telecommunications, with a specialization in mobile technologies and connected objects from the University of Technology of Troyes (UTT), Troyes, France, in 2022, where he is currently pursuing the Ph.D. degree.



Luis A. Salazar-Zendeja received the B.S. degree in mathematics from the Autonomous University of Nuevo León, San Nicolás de los Garza, Mexico, in 2015, the M.S. degree in manufacturing systems from the Tecnológico de Monterrey, Monterrey, Mexico, in 2017, and the Ph.D. degree in computer science from Centrale Lille, Lille, France, in 2022.

He currently holds a post-doctoral position at the University of Technology of Troyes, Troyes, France. His research interests include scheduling and deep learning.



Faicel Hnaïen received the engineering degree in industrial engineering from the National Engineering School of Tunis, Tunisia, in 2004, the master's degree in industrial engineering from the Polytechnic Institute of Grenoble, Grenoble, France, in 2005, and the Ph.D. degree in industrial engineering from the École des Mines de Saint-Étienne, Saint-Étienne, France, in 2008.

From 2008 to 2009, he held a temporary teaching and research position at French Institute of Advanced Mechanics, Aubière, France. Since 2009, he has been an Associate Professor at the University of Technology of Troyes (UTT), Troyes, France, and he was promoted to a Full Professor in 2024. His academic work has focused on developing optimization methods in industrial engineering.



Hichem Snoussi received the Diploma degree in electrical engineering from École supérieure d'électricité, Gif-sur-Yvette, France, in 2000, the DEA and Ph.D. degrees in signal processing from the University of Paris Sud, Orsay, France, in 2000 and 2003, respectively, and the HDR degree from the University of Technology of Compiègne, Compiègne, France, in 2009.

From 2003 to 2004, he was a Post-Doctoral Researcher with the Institut de Recherche en Communications et Cybernétique de Nantes (IRCCyN), Nantes, France. He has spent short periods as a Visiting Scientist with the Brain Science Institute, RIKEN, Wako, Japan, and Olin Neuropsychiatry Research Center, Institute of Living, Hartford, CT, USA. From 2005 to 2009, he was an Associate Professor with the University of Technology of Troyes, Troyes, France. Since 2010, he has been a Full Professor with the University of Technology of Troyes. Since 2010, he has been the in charge of CapSec Platform (Sensors for Security). From 2016 to 2020, he has been the Manager of the LM2S Laboratory, Charles Delaunay Institute, Troyes. Since 2021, he has been the Deputy Director of the LIST3N Laboratory, and the Team Leader of the M2S Group. He is the principal investigator of many research projects and industrial partnerships. His research interests include Bayesian techniques for source separation, information geometry, differential geometry, and machine learning.



Brice Nélain received the M.Sc. degree in advanced materials from Cranfield University, Wharley End, U.K., and the Engineering degree from the Ecole Centrale de Paris, Gif-sur-Yvette, France, in 2007.

From 2008 to 2023, he took over the positions of computation engineer, and an Expert Engineer at VIBRATEC, Écully, France, specialized in vibration and acoustics. In 2023, he joined the Innovation Team, VOSSLOH, Villeurbanne, France, as the Project Leader. His research was focused on ground-borne noise and vibration, mechanical fatigue design, and gearbox noises.